

1 Quick refresher on unitary transformations

Let's say we have some Hamiltonian $\hat{H}$, which we examine in a basis $\{|v_j\rangle\}$ with $j = 1, \dots, \dim(\mathcal{H})$ and $\hat{H}: \mathcal{H} \rightarrow \mathcal{H}$. The matrix representation of the operator $\hat{H}$ in this basis is thus given by

$$H_{ij} = \left\langle v_i \left| \hat{H} v_j \right. \right\rangle = \langle v_i | \hat{H} | v_j \rangle. \quad (1)$$

Applying a unitary transformation U (with $U^\dagger U = \mathbb{1} = UU^\dagger$), i.e.

$$|v_j\rangle \rightarrow |\tilde{v}_j\rangle = U |v_j\rangle \quad \& \quad \hat{H} \rightarrow \hat{\tilde{H}} = U \hat{H} U^\dagger, \quad (2)$$

yields the matrix elements

$$\tilde{H}_{ij} = \langle \tilde{v}_i | \hat{\tilde{H}} | \tilde{v}_j \rangle = \langle v_i | \underbrace{U^\dagger U}_{=\mathbb{1}} \hat{H} \underbrace{U^\dagger U}_{=\mathbb{1}} | v_j \rangle = \langle v_i | \hat{H} | v_j \rangle = H_{ij}. \quad (3)$$

Hence the matrix representation of an operator does *not* change under a unitary transformation.